0.1 MC Samples Used

The main MC samples used in the analysis are as follows,

- 200 M $K \rightarrow \pi^0_D \pi^+$
- 50 M $K \rightarrow \pi^0_D \mu^+ \nu_\mu$

0.2 Event Selection

- Require either the Control Trigger (sometimes described as Minimum Bias Trigger), Multi Track (MT) or Electron Multi Track (eMT) which are described in Chapter ??
- A vertex was selected if the total momentum was below $78/, GeV$, a total positive charge of 1, a χ^2 less than 40 and a vertex position between 105 and 180 m.
- The tracks associated to the vertex are selected if they are in acceptance of all four Straw spectrometer chambers and the CHOD. The track momentum must be greater than $5\,GeV$ and less than $60\,GeV$ with a $^+$ of less than 30. The tracks must traverse within 20 cm of each other.
- Require the presence of a three track in the vertex with a selected positive charged pion (π^+), and a selected electron positron pair ($\pi^\pm$). The pion is selected using missing mass between the Kaon momentum which is taken from the average beam momentum and the two positive tracks. If the positive track is within 5 sigma of the π^0 mass hypothesis then it is selected as a charged pion. If both positive tracks satisfy the condition the event is vetoed. From

MC only approximately $0.1/, \%$ of Dalitz decays have both positive tracks satisfying the selection of the charged pion. The electron was assigned to the only negative track and the negative track which doesn't pass the charged pion selection was selected as the positron.

- A selected photon is also required. The missing momentum of the vertex was calculated with the Kaon momentum and the total vertex momentum and was used to extrapolate a straight line from the vertex position to the z plane of the LKr calorimeter. Within a radius of 15 cm of the extrapolated position a single cluster with a minimum energy of $5\,GeV$ is selected as a photon from the decay. The photon energy has to satisfy the condition of being within $5\,GeV$ of the calculated missing vertex momentum.
- An elliptical cut is used to reduce background that would overwise pass the selection. The reconstructed mass of the electron, positron and photon is compared against the reconstructed mass of the electron, positron, photon and charged pion.

The cut on the reconstructed missing mass of the kaon, charge pion system was calculated using the Dalitz MC sample and is shown in Figure 1. The sigma is taken from the fitted guassian on the plot and calculated as,

$$\sigma = 3.984 \pm 0.007 \tag{0.1}$$

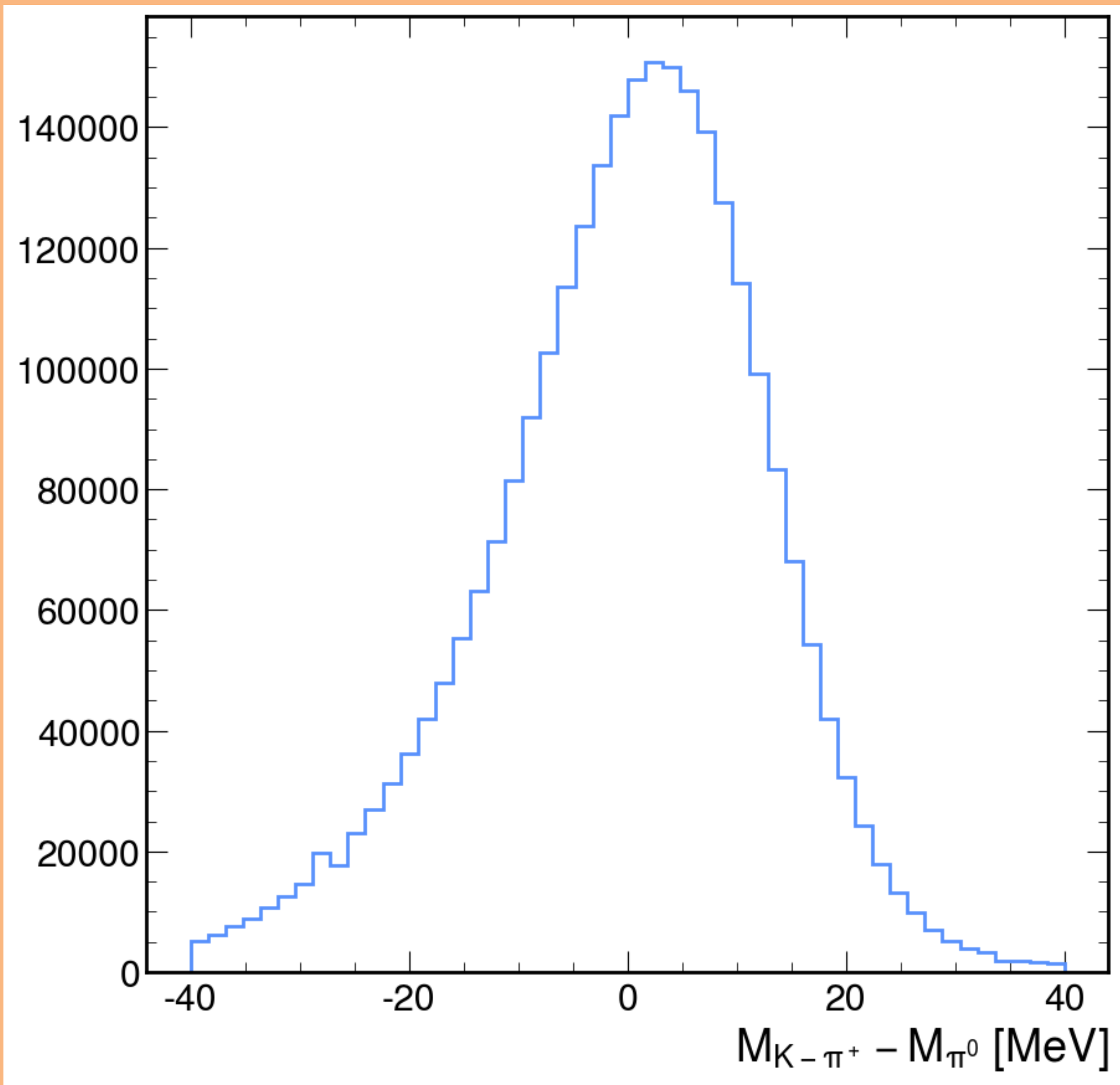

Figure 1: Reconstructed missing mass of Kaon and charged pion compared to the known neutral pion mass

The elliptical cut on the final reconstructed masses was defined using the Dalitz MC sample to collect the maximum amount of events passing the selection while making the cut a tight as possible. The simgas for the reconstructed masses $M_{ee\gamma}$ and $M_{\pi ee\gamma}$ is calculated from fitting the distributions shown in Figure 2. The calculated sigma for the reconstructed mass $M_{\pi ee\gamma}$ was calculated as

$$\sigma_{M\pi ee\gamma} = 2.47 \pm 0.004 \tag{0.2}$$

and for $M_{ee\gamma}$,

$$\sigma_{M_{ee\gamma}} = 1.329 \pm 0.009 \tag{0.3}$$

which were used as a origin for the elliptical parameters for the cut.

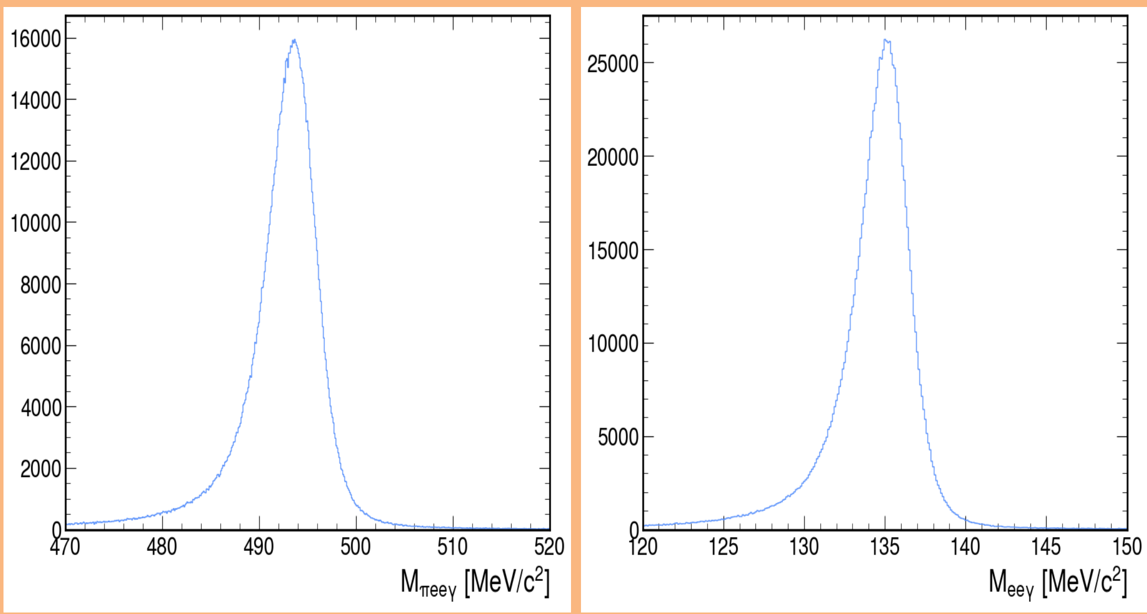

Figure 2: Reconstructed masses of selected particles, left the total of charged pion, electron, positron and gamma and on right, electron, positron and gamma.

The equation used for the elliptical cut in the selection is defined below,

$$\frac{((x - x_c)cos(\theta) + (y - y_c)sin(\theta))^2}{a^2} + \frac{((y - y_c)cos(\theta) - (x - x_c)sin(\theta))^2}{a^2} <= 1 \tag{0.4}$$

where x_c and y_c is the centre of the ellipse in the x and y dimensions. θ being the rotation of the ellipse, a and b are the length of the semi-major and the semi-minor axis respectively. x and y are the values from the selection and the event is selected if the equation returns a value that is less than or equal to 1. After first starting with the results from Figure 2 and optimizing for a tight cut with a acceptable number of events passing produced parameters below,

- ($x_c = 493.37$): The x-coordinate of the ellipse's center.
- ($y_c = 134.89$): The y-coordinate of the ellipse's center.

- ($a = 10.24$): The semi-major axis of the ellipse ($4 \times \sigma_{M\pi ee\gamma}$).
- ($b = 4.29$): The semi-minor axis of the ellipse ($3 \times \sigma_{Mee\gamma}$).
- ($\theta = 0.4887$): The rotation angle of the ellipse in radians.
- ((x, y)): The point being tested for inclusion within the ellipse.

Which produced an ellipse used in the selection which is shown in 3

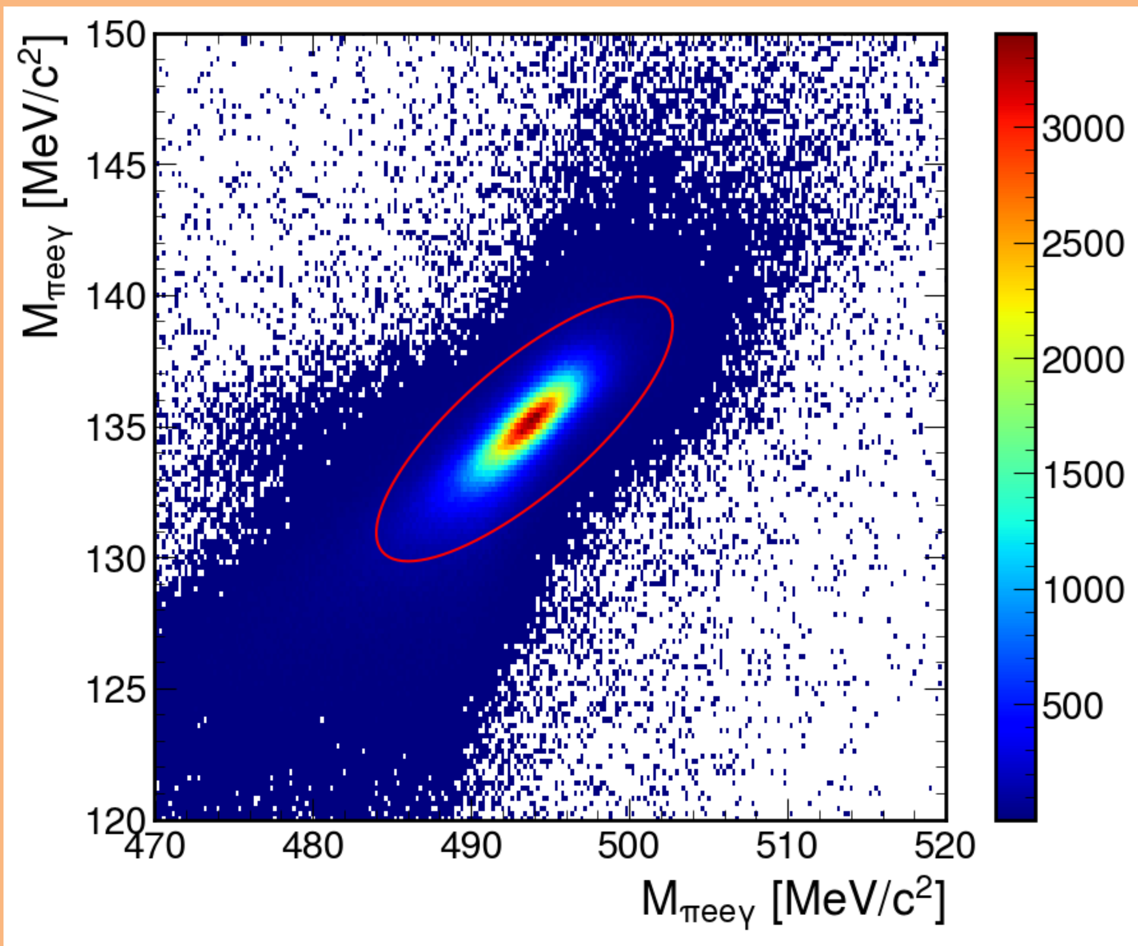

Figure 3: Reconstructed missing mass of $M_{\pi ee\gamma}$ compared against $M_{ee\gamma}$ produced from a K2pi Dalitz MC sample, with the defined elliptical cut used in the selection

The acceptance of the Dailtz sample is defined across the reconstructed electron position mass in the range $2m_e < M_e e < m_{\pi^0}$ and is compared against the acceptance of the NA48/2 analysis [?] in Figure 4. Improving the selection using kinematic cuts compared to the calorimeter PID improved the acceptance at high $M_e e$ compared to NA48/2. It was expected that NA48/2 would have an acceptance advantage due to its design as a shorter experiment. If the same analysis procedure was employed using calorimetric PID the acceptance would not be competitive with NA48/2

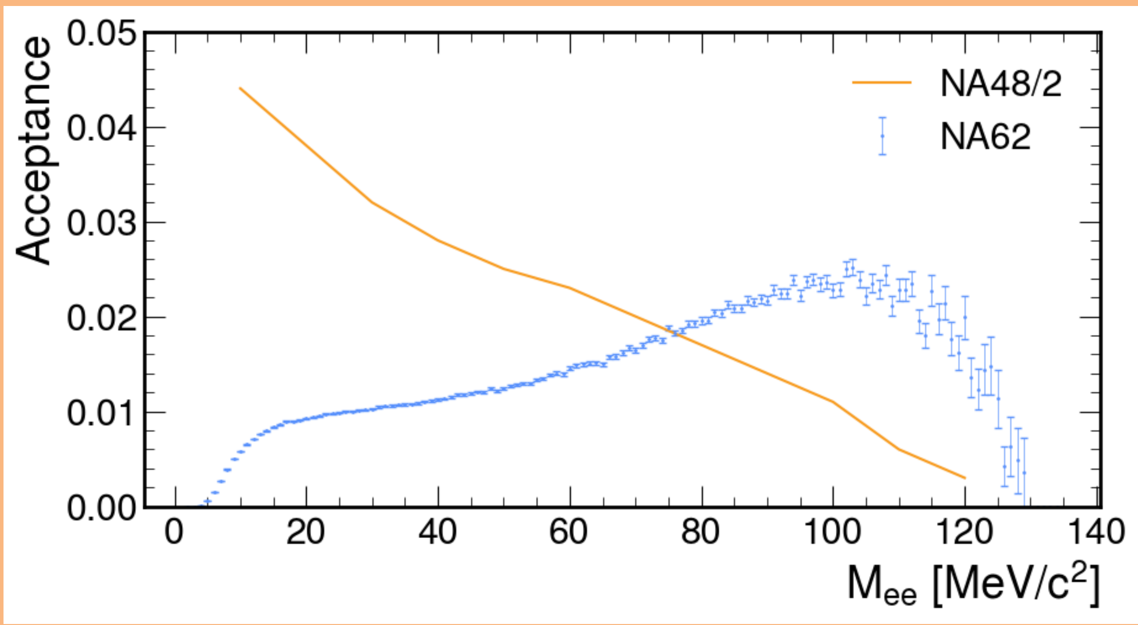

Figure 4: Acceptance of the NA62 selection compared against the acceptance produced in the NA48/2 analysis [?]

0.3 Integrated Kaon Flux

sec:kaonFlux The total number of data events passing the selection for 2022 was 22,059,350 with the number of K^+ decays within the fiducial volume for the 2022 dataset was computed as

$$N_K^{2022} = \frac{N_{selected}}{(A_{2\pi D}B_{2\pi D} + A_{\mu 3D}B_{\mu 3D})} = (1.65 \pm 0.05) \times 10^{12} \qquad (0.5)$$

Where $N_{selected}$ is the number of data events passing the selection, $A_{2\pi D}$ and $A_{\mu 3D}$ are the acceptances of the selection and are evaluated using MC samples and $B_{2\pi D}$,

$B_{\mu 3D}$ are the branching fractions of the decays [?]. As the MC samples employ the trigger emulators provided by the NA62 Framework the trigger efficiencies for both L1 and HLT are accounted for within the acceptances that are defined in Table ??.

MC Sample	**Acceptance**
$K_{2\pi D}$	0.56%
$K_{\mu 3D}$	0.0027%

Table 1: Acceptances of $K_{2\pi D}$ and $K_{\mu 3D}$ samples for the $K_{2\pi D}$ selection

The 2022 sample was available when the MC background method was employed. However when the second and final background method was finalised the 2023 dataset was available so it was used. The integrated number of Kaons was then calculated using the 66,839,882 data events passing the selection for 2022+2023

$$N_K^{2022/23} = (4.95 \pm 0.15) \times 10^{12} \tag{0.6}$$

The large increase in data events passing the selection when the 2023 dataset was added was due to the change in downscaling for the eMT trigger which went from $3 \rightarrow 1$. The reconstructed masses for data and MC passing the $K_{2\pi D}$ selection for 2022+2023 are shown in Figure 5. The MC samples have been scaled using the calculated N_K. The plots for the 2022 sample produce similar results but with lower statistics.

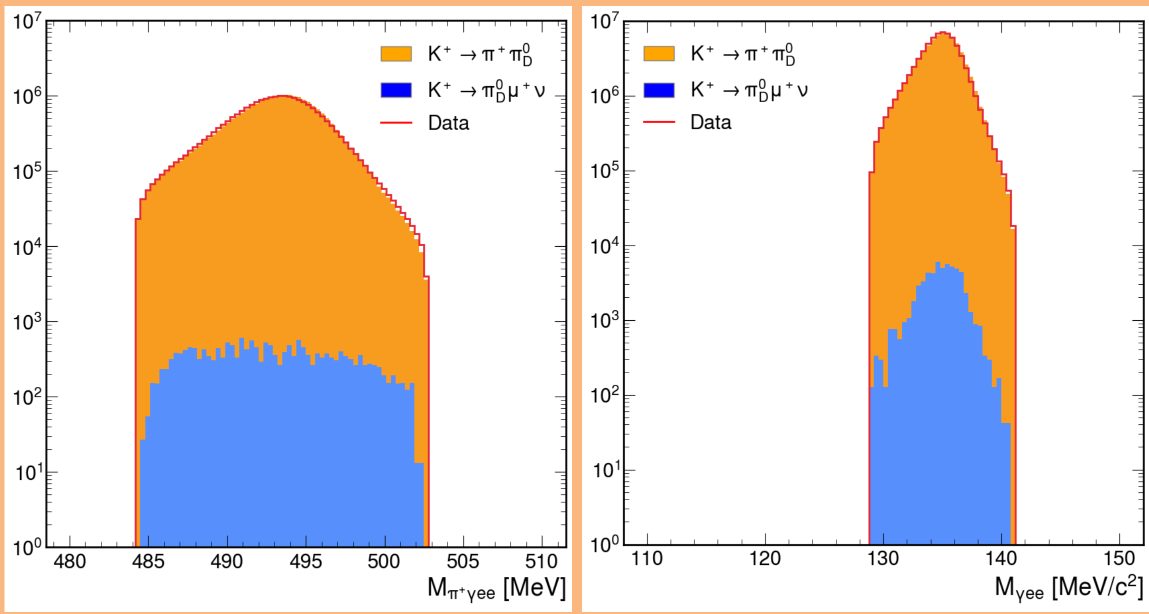

Figure 5: Reconstructed masses $K_{2\pi D}$ and $K_{\mu 3D}$ for MC and data

0.4 Dark Photon Search

Two methods were employed in the background estimation for the Dark Photon Search. First used generated MC samples of the expected background as the background estimate, however as described below produced unsatisfactory results, which led on to the second method, using a data driven method for the background estimation. Both methods are described below

0.4.1 MC Background Method

As stated before, the MC method only used the 2022 dataset. It was expected that the limited size of the MC samples could impact the final result when used for the background estimate, however it was not feasible to directly increase the statistics on the main samples due to compute time and file size. It was possible to increase the statistics across a portion of the reconstructed mass spectra using a feature of the Framework. It is possible for the true invariant mass of the electron positron pair to be used to calculate the kinematic variable x from Equation 0.7.

$$x = \frac{(Q_{e^-} + Q_{e^+})^2}{m_{\pi^0}^2} = \frac{m_{ee}^2}{m_{\pi^0}^2} \tag{0.7}$$

The resultant distribution from the MC sample is shown in Figure 6. The decay width in Equation 0.8 is strongly dependent on this kinematic term, reducing to zero as the reconstructed invariant mass reached the π^0 mass. The Framework allows for the simulation of the Dalitz decay where the kinematic variable $x > 0.5$. From the true MC distribution in Figure 6 the fraction of events above 0.5 was 0.003. Producing a simulation with the enabled, even with a fraction of size of the main sample would decrease the errors on the MC counts by orders of magnitude.

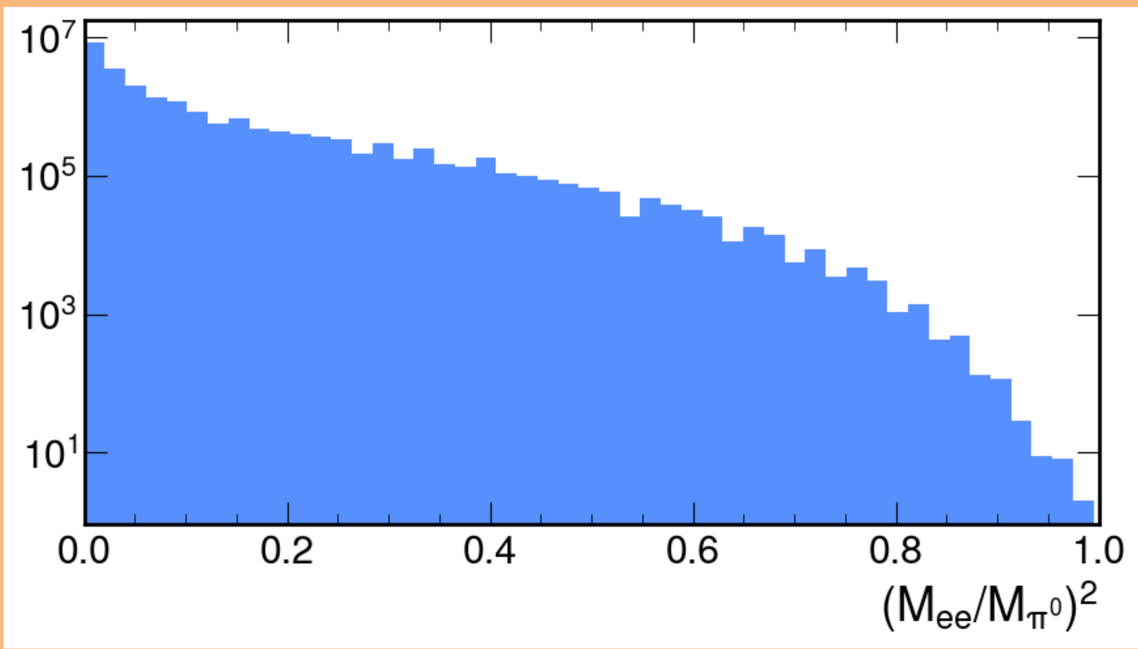

Figure 6: Reconstructed missing mass of electron positron pair of in terms of reconstructed kinematic variable x

For the rest of the study the following MC samples were used

- 200 M $K \rightarrow \pi^0_D \pi^+$ $(x < 0.5)$

- 5 M $K \to \pi_D^0 \pi^+$ $(x > 0.5)$
- 50 M $K \to \pi_D^0 \mu^+ \nu_\mu$

In order to improve the combination of the two $K_{2\pi_D}$ samples the large sample had events where $x > 0.5$ removed to allow for the new sample to cover that region. After the full selection on the MC samples and data, the reconstructed mass of the electron positron pair is shown in Figure 7.

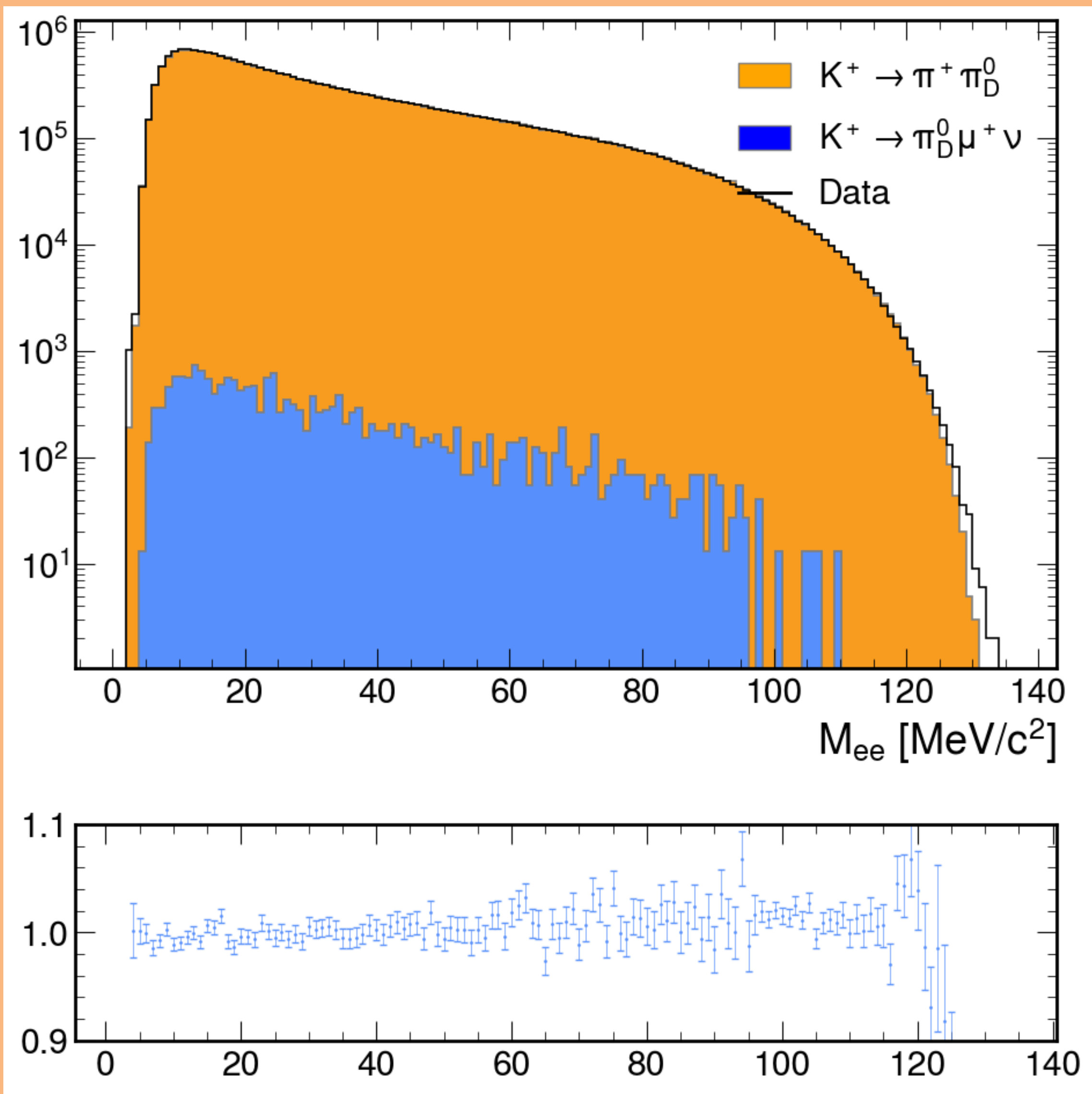

Figure 7: Reconstructed mass of electron positron pair of data and MC samples.

The samples used include the simulation of the π_D^0 using the lowest-order differential decay rate [?] and is defined below

$$\frac{d^2\Gamma}{dxdy} = \Gamma_0 \frac{\alpha}{\pi} |F(x)|^2 \frac{(1-x)^3}{4x} \left(1 + y^2 + \frac{r^2}{x}\right) \tag{0.8}$$

where Γ_0 is the $\pi^0 \to \gamma\gamma$ differential decay rate, α is the fine-structure constant, F(x) is the pion transition form factor (TFF) and x,y are kinematic variables defined below

$$y = \frac{2Q_{\pi^+}(Q_{e^-} - Q_{e^+})}{m_{\pi^0}^2(1-x)} \tag{0.9}$$

where Q_{e^-}, Q_{e^+} and Q_{π^+} are the four-momentum of the electron, position and charged pion. The TFF is parametrised as $F(x) = 1 + ax$, with a being a parameter set by experiments or theory. ADD THEORY BIT. The Particle Data Group (PDG) average for all experiments was $a = 0.03350.0031$ [?]. NEED MORE DESCRIPTION WHY CANT USE THEORY OR EXPERIMENT. The background estimation using the MC cannot rely on the input of theory or experiments due to the underlying uncertainties. In the NA62 Framework $a = 0.032$ so a correction had to be made to remove the applied TFF and re-weight using the events passing the selection. It is clear that in Figure 8 which shows the data over MC from Figure 7 but rather than in terms of M_{ee} plot against x from Equation 0.7, suggests that there is a discrepancy. MC and data no longer match but is dependent along x suggesting that the TFF found in the Framework is not acceptable for use in the background estimate.

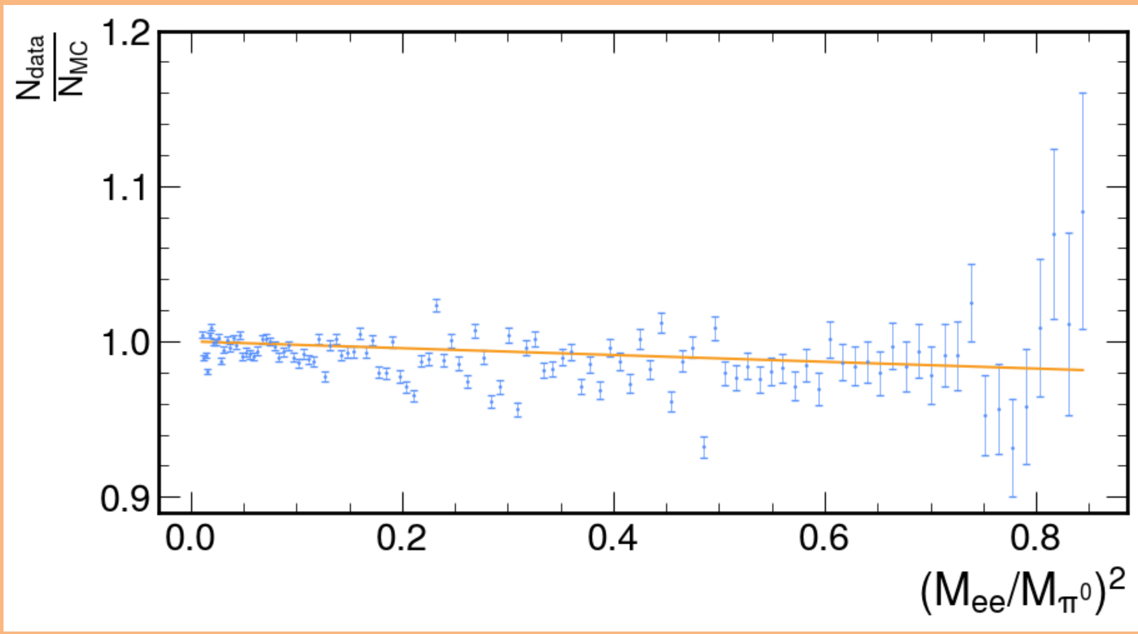

Figure 8: Data vs MC in terms of reconstructed kinematic variable x with the framework transition form factor

The effective TFF of the analysis can be found by removing the Framework value by inverting the effects of the square of the TFF,

$$F(x)^2 = (1 + ax)^2 \tag{0.10}$$

which resulted in the distribution shown in Figure 9. The MC with the removed effects of the TFF was fitted with equation 0.10 using a least χ^2 method, with the resultant line shown in Figure 9, representing a value for a

$$a = 0.0210 \pm 0.0029 \tag{0.11}$$

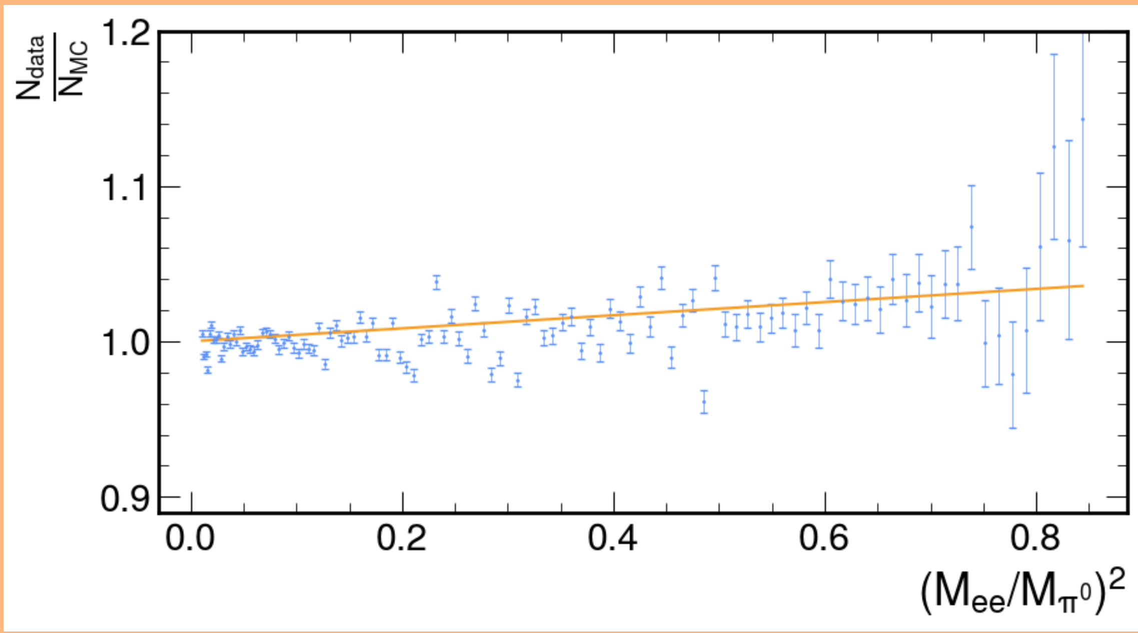

Figure 9: Data vs MC in terms of reconstructed kinematic variable x with the framework transition form factor removed

The correction resulted in slightly better Data/MC agreement shown in Figure 10 when compared to Figure 7.

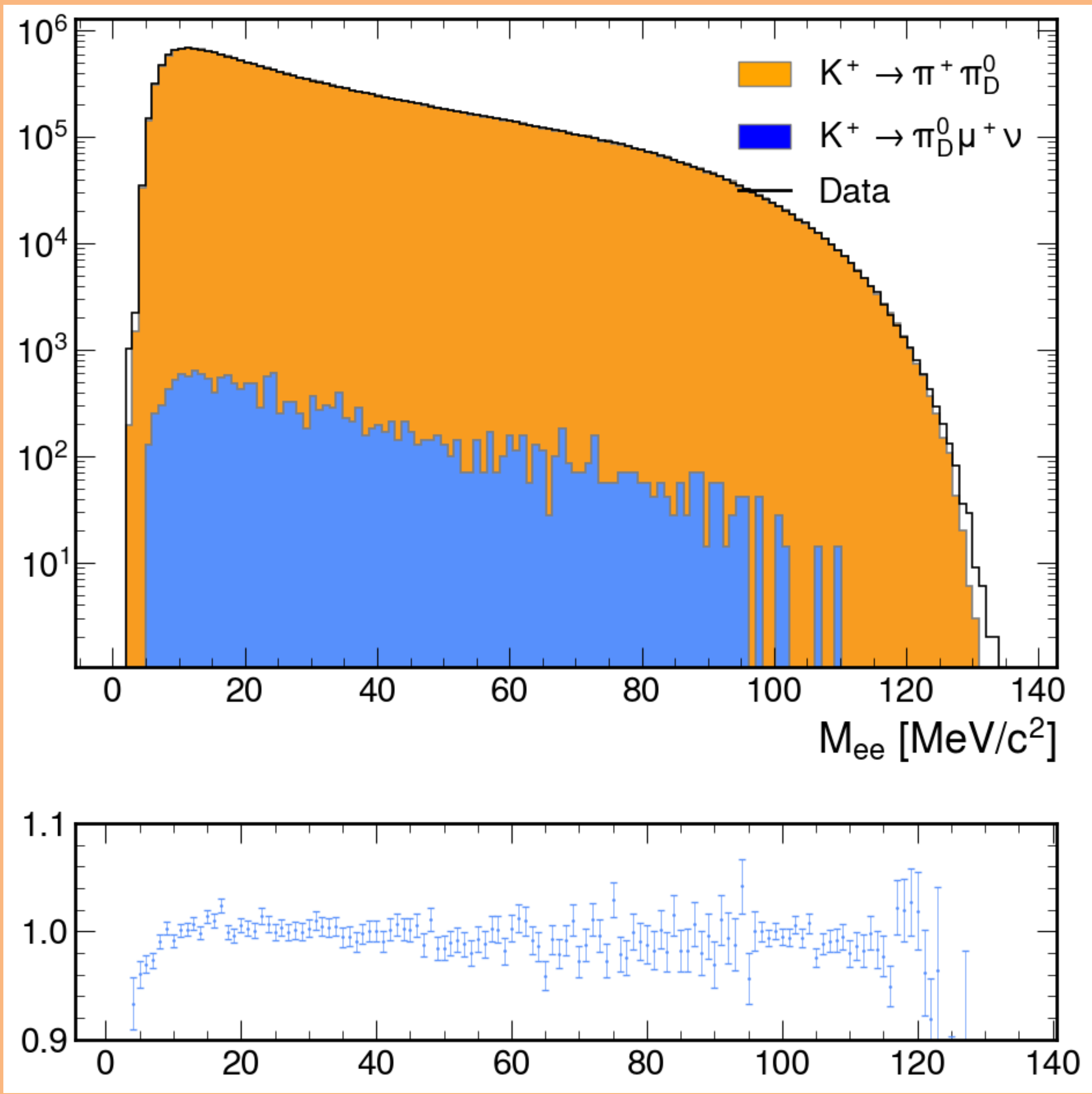

Figure 10: Reconstructed mass of electron positron pair of data and MC samples corrected for the effective transition form factor.

The final counts for events passing the selection in terms of the reconstructed invariant mass for each MC sample is shown in Figure 11, where it is clear where the two $K_{2\pi_D}$ samples have been combined.

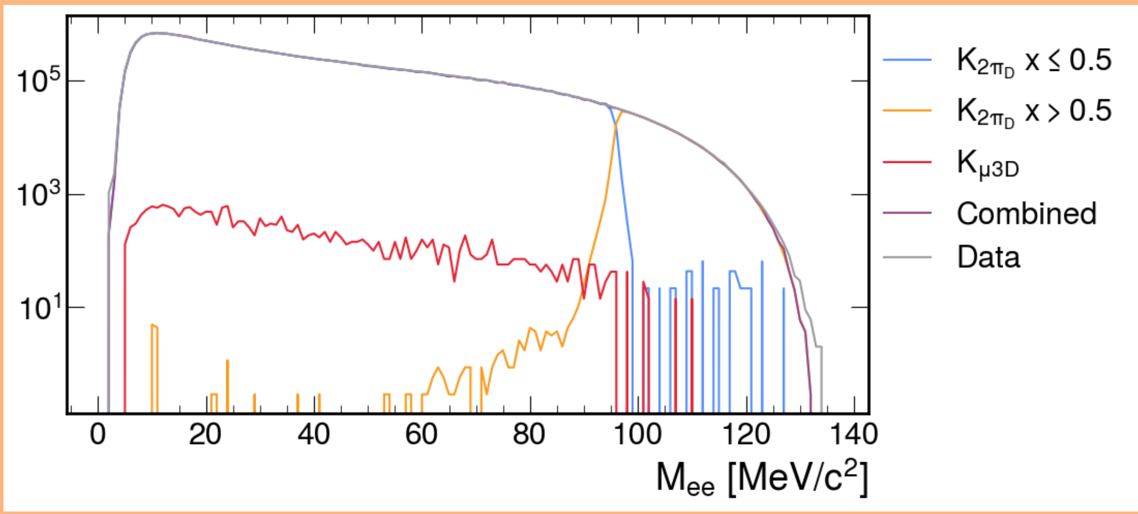

Figure 11: Reconstructed mass of electron positron pair of data, individual MC samples and combined MC.

The resultant statistical errors were calculated for each individual MC sample and combined to produce a combined MC error for each mass bin and is shown in Figure 12. It was clear that implementing the biased $K_{2\pi_D}$, significantly reduced the statistical error of the combined sample by an order of magnitude compared to using only the original sample.

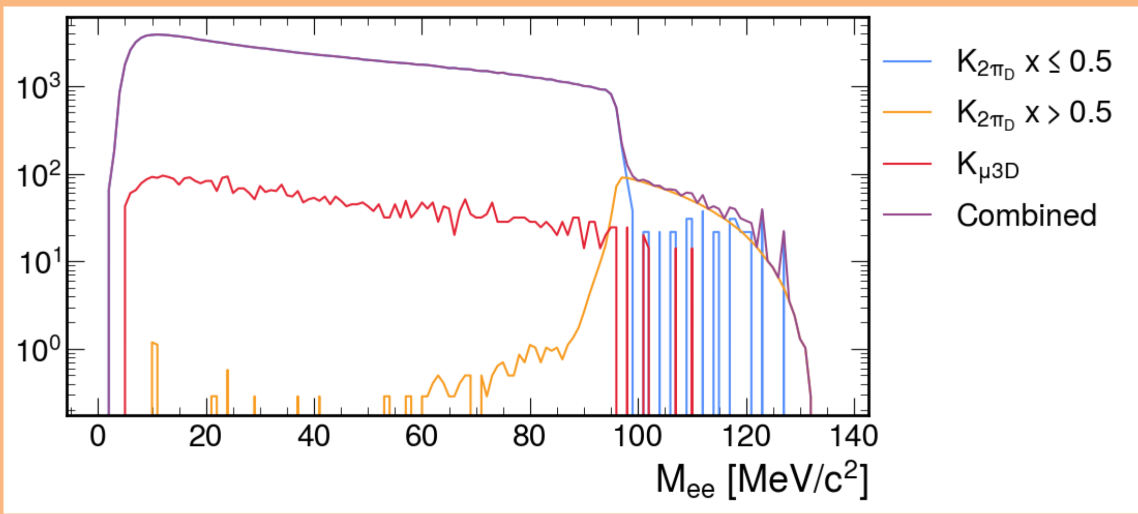

Figure 12: Statistical error of MC samples in terms of reconstructed mass of electron positron pair.

The range of the Dark Photon search was defined as $9 \leq m_{A'} \leq 120 MeV/c^2$. The lower mass boundary is defined due to the falling geometric acceptance of the decay and the low electron identification efficiency at low momentum. The upper boundary was also defined due to both the falling geometrical acceptance and suppressed decay width. To select the search window centers and the size of the windows, the reconstructed mass resolution was considered. The reconstructed mass resolution was calculated using MC where the difference between the true mass and the reconstructed mass of the electron positron pair was collected for each mass bin and is shown in Figure 13.

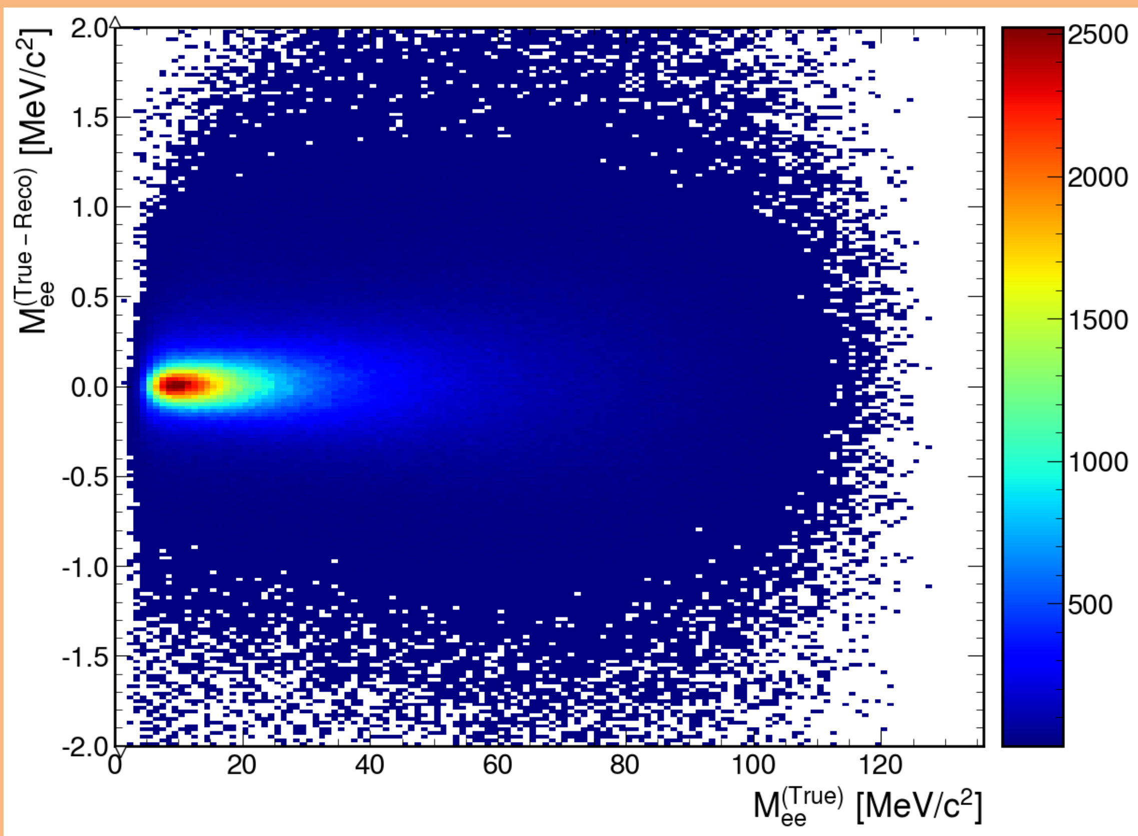

Figure 13: True invariant mass minus reconstructed invariant mass of the electron positron pair in terms of true invariant mass.

For each invariant mass bin the Y projection is taken and a gaussian fit is conducted, where the sigma of the fit is recorded to produce the plot shown in Figure 14. The resultant points in blue are fitted to a straight line shown in orange, producing an equation for the reconstructed invariant mass resolution,

$$\sigma_m(m_{ee}) = 0.03755 MeV/c^2 + 0.00554 \times m_{ee} \quad (0.12)$$

The NA62 resolution is a significant improvement compared to NA48/2, shown as red line in Figure 14 due to improved detector design and setup. Which results in smaller search mass windows compared to the previous study.

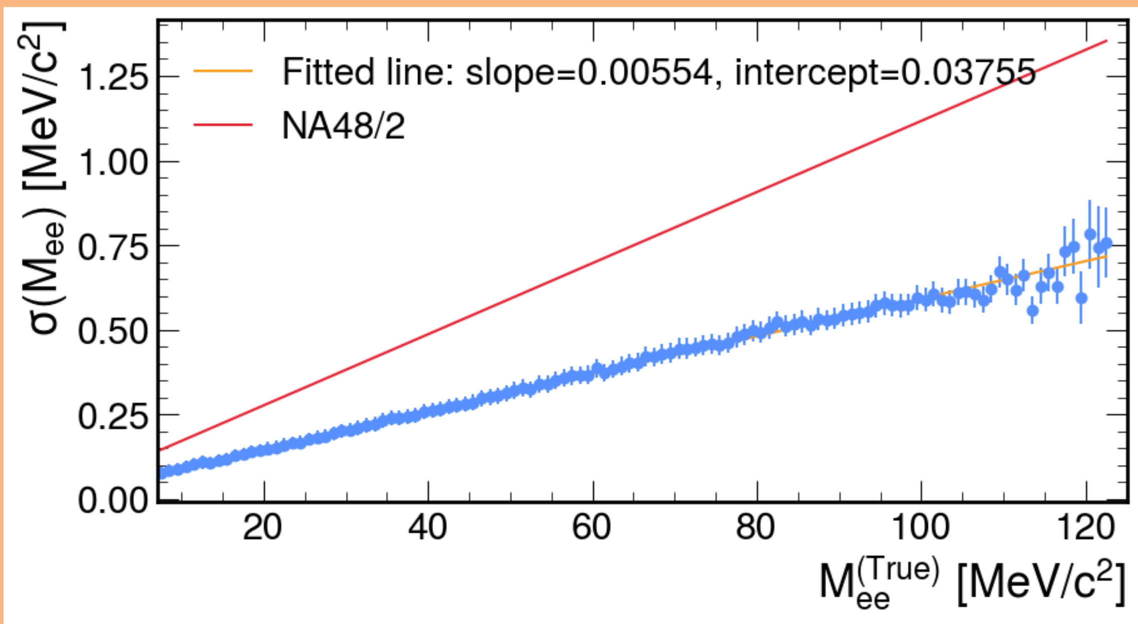

Figure 14: Sigma of the difference between true and reconstructed invariant mass of the electron-positron pair in bin of true invariant mass.

Using Equation 0.12 the window centres are defined as being separated by $3\sigma_m(m_{A'})$, with a window width of $\pm 1.5\sigma_m(m_{A'})$ rounded to the nearest 0.01 MeV/c^2 which was the size of the saved bins of the invariant mass spectrum, resulting in 177 mass values tested.

For each DP mass value the number of data events N_{obs} was compared to the number of expected events obtained from MC N_{exp}. The N_{obs}, N_{exp} and their subsequent errors are shown in Figure 15. The error on the number of data events δN_{obs} resulted from the statistical error from the number of events within each bin $\delta N_{obs} = \sqrt{N_{obs}}$. Similar, the error on the expected number of events was due to the statistical error on the number of MC events within the search window, but had to be corrected for the scale factor imposed on the MC to equate for the correct effective kaon flux, defined in Section **??**, resulting in $\delta N_{exp} = sf \times \sqrt{N_{exp}/sf}$ where sf was the scale factor used to scale the MC sample. It would of been require to include statistical errors due to the trigger efficiencies but as mentioned previously all MC samples employed a trigger emulator so this was not required.

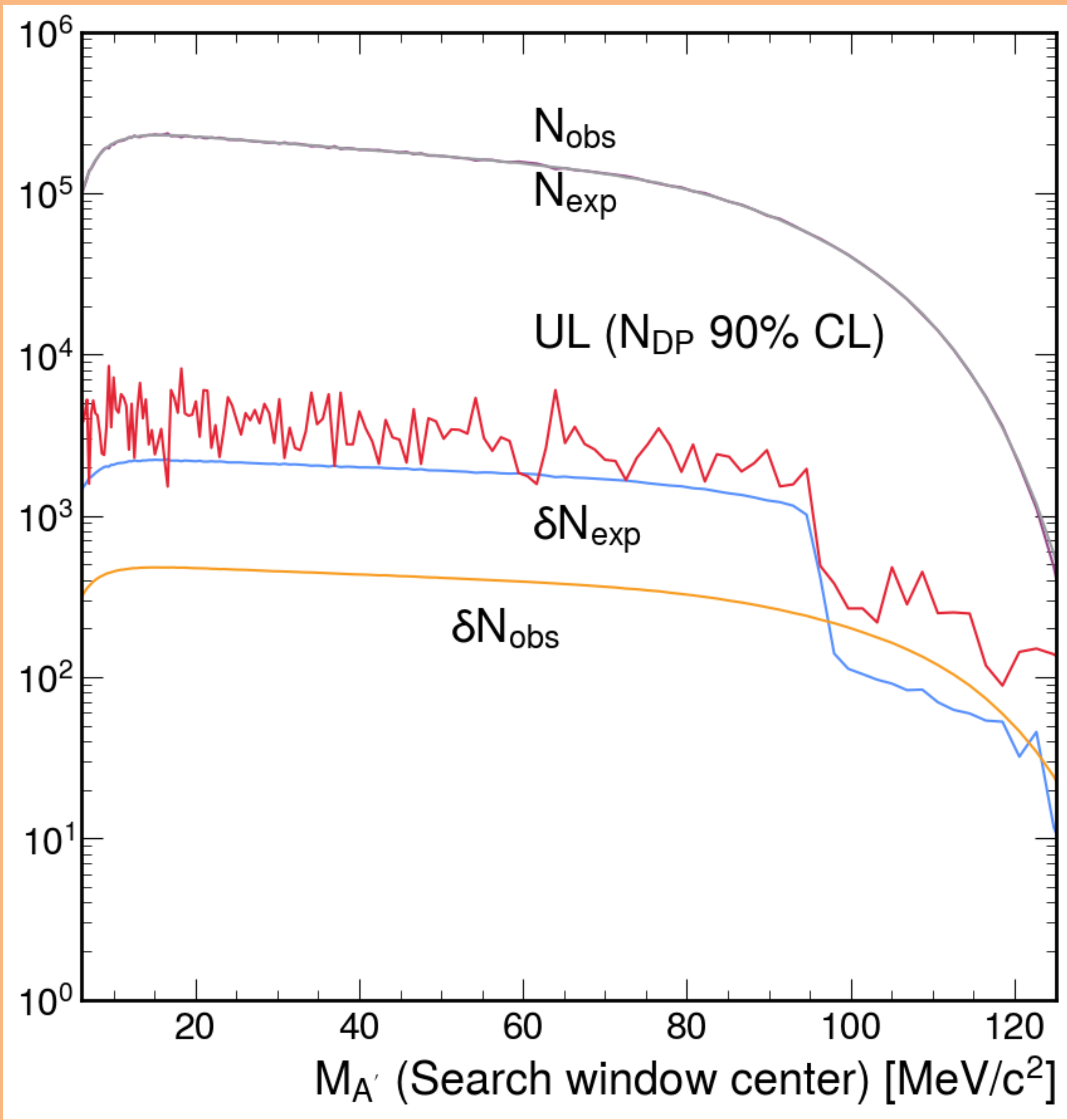

Figure 15: Number of observed data events (N_{obs}), expected number of events from MC (N_{exp}), the respective errors (δN_{obs} and δN_{exp}) and the resultant upper limit on the number of Dark Photon candidates to a 90% confidence level.

For each DP mass window the local significance of the DP signal was calculated,

$$Z = \frac{N_{obs} - N_{exp}}{\sqrt{\delta N_{obs} + \delta N_{exp}}} \tag{0.13}$$

and is shown in Figure 16. Across the search window the local significance is never greater than 3σ resulting in no DP signal observed. The confidence intervals for the number of DP decays ($A' \to e^+e^-$) at a 90% CL for each DP mass value was calculated using N_{obs}, N_{exp} and δN_{exp} using the CLs method recommended by the Particle Data Group [?] and employs the method setout in [?].(MAYBE DESCRIBE METHOD) The resultant upper limits on the number of DP candidates (N_{DP}) at a 90% CL is shown in Figure 15.

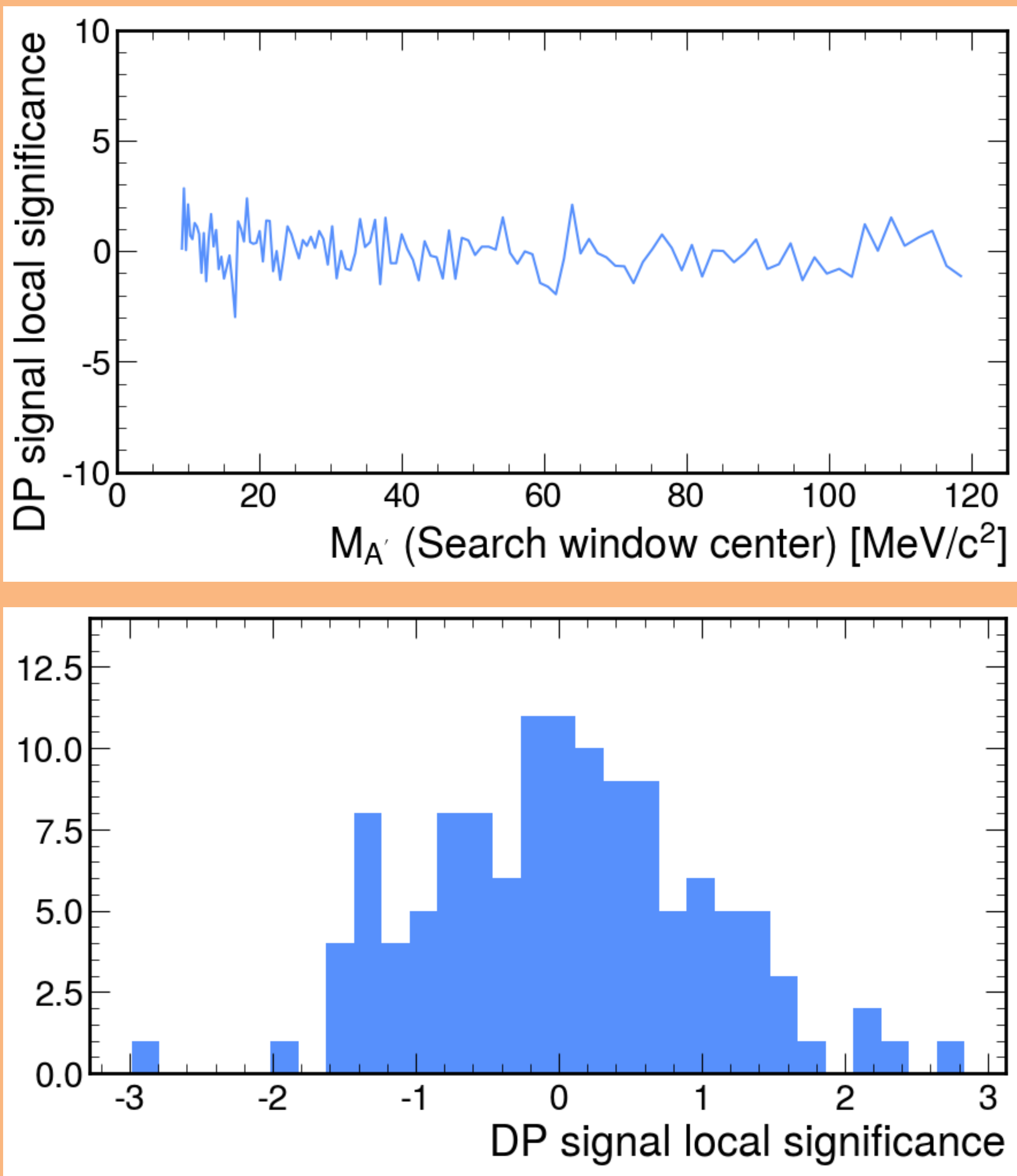

Figure 16: Local significance of DP signal across the search window $9 \leq m_{A'} \leq 120 MeV/c^2$. Top, in terms of DP mass and bottom, the y projection of the top plot.

The UL on the N_{DP} was used to calculate the resultant 90% CL on the branching fraction of the decay $\pi^0 \to \gamma A'$, which was computed with Equation 0.14, with the assumption that $\mathcal{B}(A' \to e^+e^-) = 1$.

$$\mathcal{B}\pi \to \gamma A = \frac{N_{DP}}{N_K Br_{K_{2\pi}} A_{DP}(K_{2\pi})} \tag{0.14}$$

where N_K was the effective kaon flux defined in Section sec:kaonFlux, $Br_{K_{2\pi}}$ is the branching fraction of $K^+ \to \pi^+\pi^0$ [?] and $A_{DP}(K_{2\pi})$ which was the acceptance of the DP signal. $A_{DP}(K_{2\pi})$ was calculated using the acceptance of the π^0_D decay shown in Figure 4, parametrized with a polynomial to allow for the calculation of the acceptance in terms of DP mass. The resultant branching fraction at a 90% across the DP mass range is shown in Figure 17. The dramatic reduction of the UL at about 90 MeV/c^2 was due to the improved statistical errors provided by using the second $K_{2\pi_D}$ sample biased using the kinematic variable. The red dashed lines provide the range at which the UL of the the branching fraction for NA48/2 result [?], suggesting with improved statistical the possibility of a two orders of magnitude improvement.

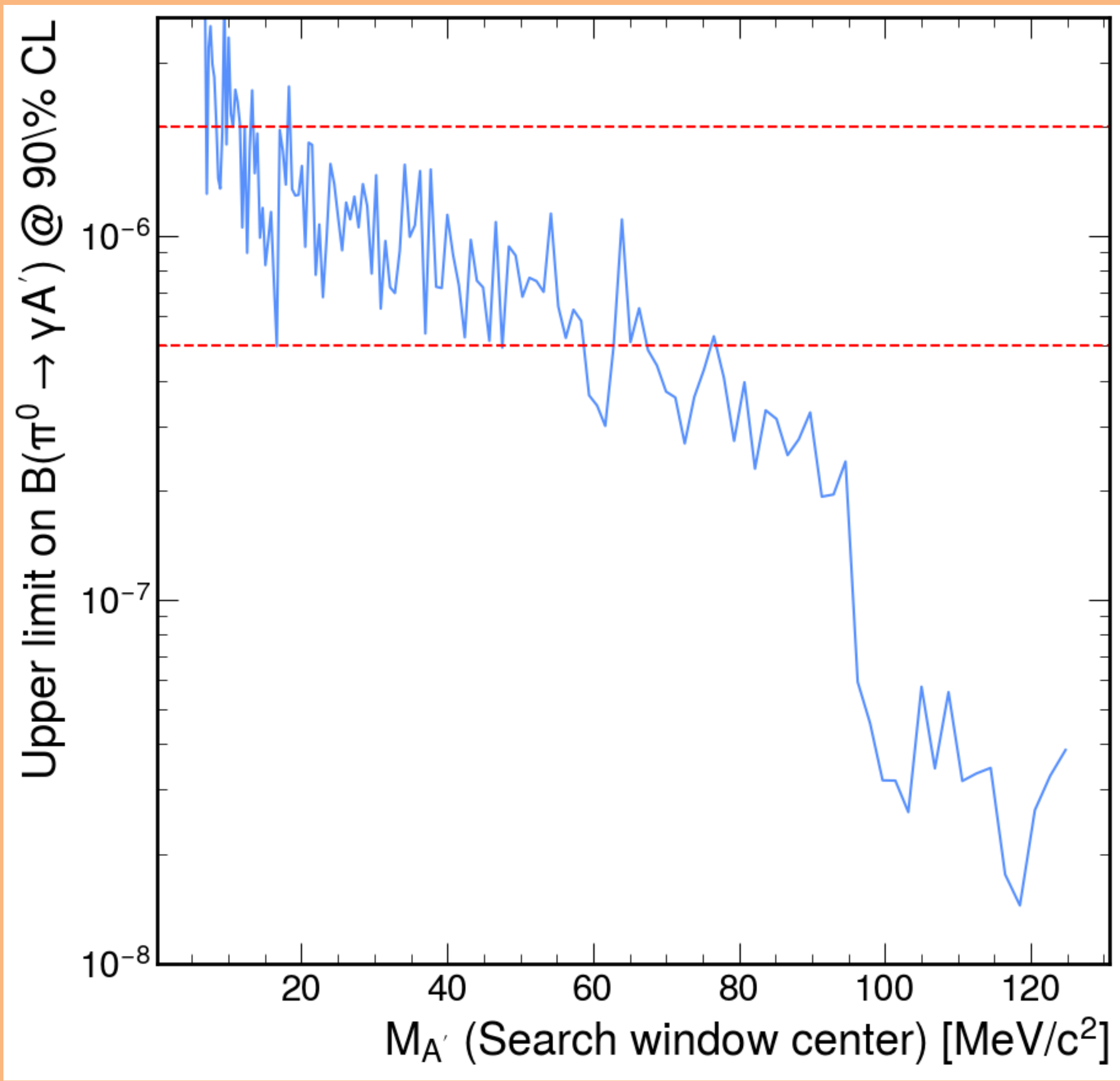

Figure 17: Upper limit on the branching fraction of $\pi^0 \to \gamma A'$ at a 90% CL for each Dark Photon mass value.

The UL on the branching fraction was used to to set the upper limits on the mixing parameter ϵ^2 for each DP mass value. ϵ^2 was calculated from (NEED TO PUT EQUATION FROM THEORY) resulting in Equation 0.15.

$$\epsilon^2 = \frac{B(\pi^0 \to \gamma A')}{2\left(1 - \frac{m_{A'}^2}{m_{\pi^0}^2}\right)^3 B(\pi^0 \to \gamma\gamma)} \tag{0.15}$$

The resultant UL of the mixing parameter ϵ^2 at a 90% CL across the DP mass range is show in Figure 18. It is clear that the NA62 result for the 2022 data was not competitive at low invariant mass ranges up to around $85MeV/c^2$, where the statistics improve due to the biased sample. Suggesting that a low mass ranges the UL is limited by the statistical error on the MC sample. It is expected if the sample was at the same statistical level as the biased sample the UL could be improved by at least an order of magnitude.

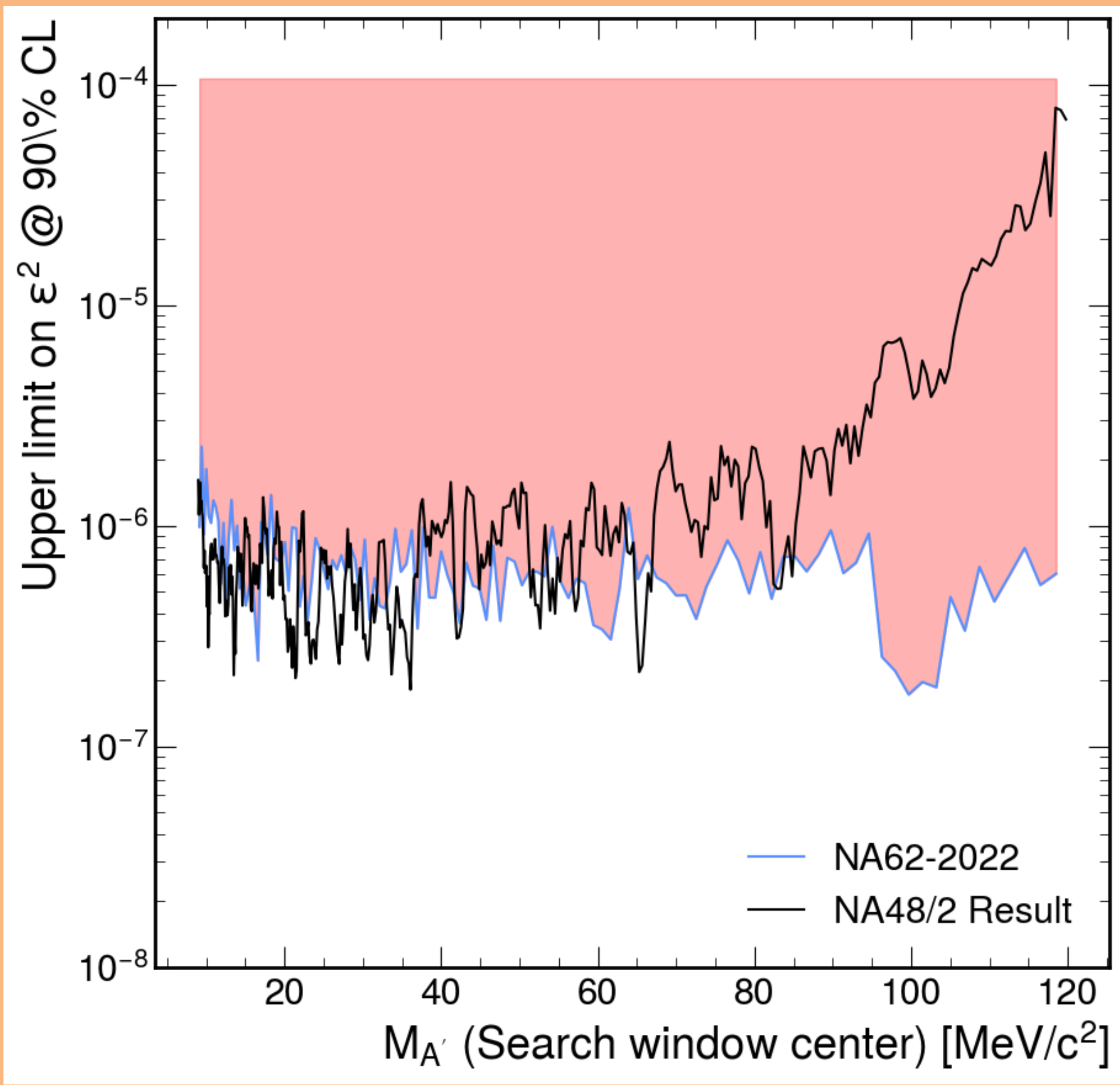

Figure 18: Upper limit mixing parameter ϵ^2 at a 90% CL for each Dark Photon mass value with the result from NA48/2 for comparison [?].

It is clear that without increased MC statistics, the study would not be competitive with the already published result from NA48/2 [?] except for high mass ranges above $85MeV/c^2$. A new method could be employed that removed the statistical error introduced by the MC for the background estimation.

0.4.2 Data Driven Background Method

The data driven background method employed the same selection setout in the previous section but rather than using the MC samples for the background estimation the data itself was used. Eventually the 2023 dataset was implemented along with the 2022. The number of data events passing the selection for the 2022/23 datasets was 66,839,882, which is just over a 3 fold increase compared to just the 2022, even though both years had similar run durations. This was due to a change in the defined downscaling for the eMT trigger line which was reduced from 3 to 1, resulting in a total effective N_K calculated with Equation 0.6,

$$N_K^{2022/23} = (4.95 \pm 0.15) \times 10^{12} \tag{0.16}$$

The resultant reconstructed invariant mass of the electron-positron pair for events passing the selection is Shown in Figure 19.

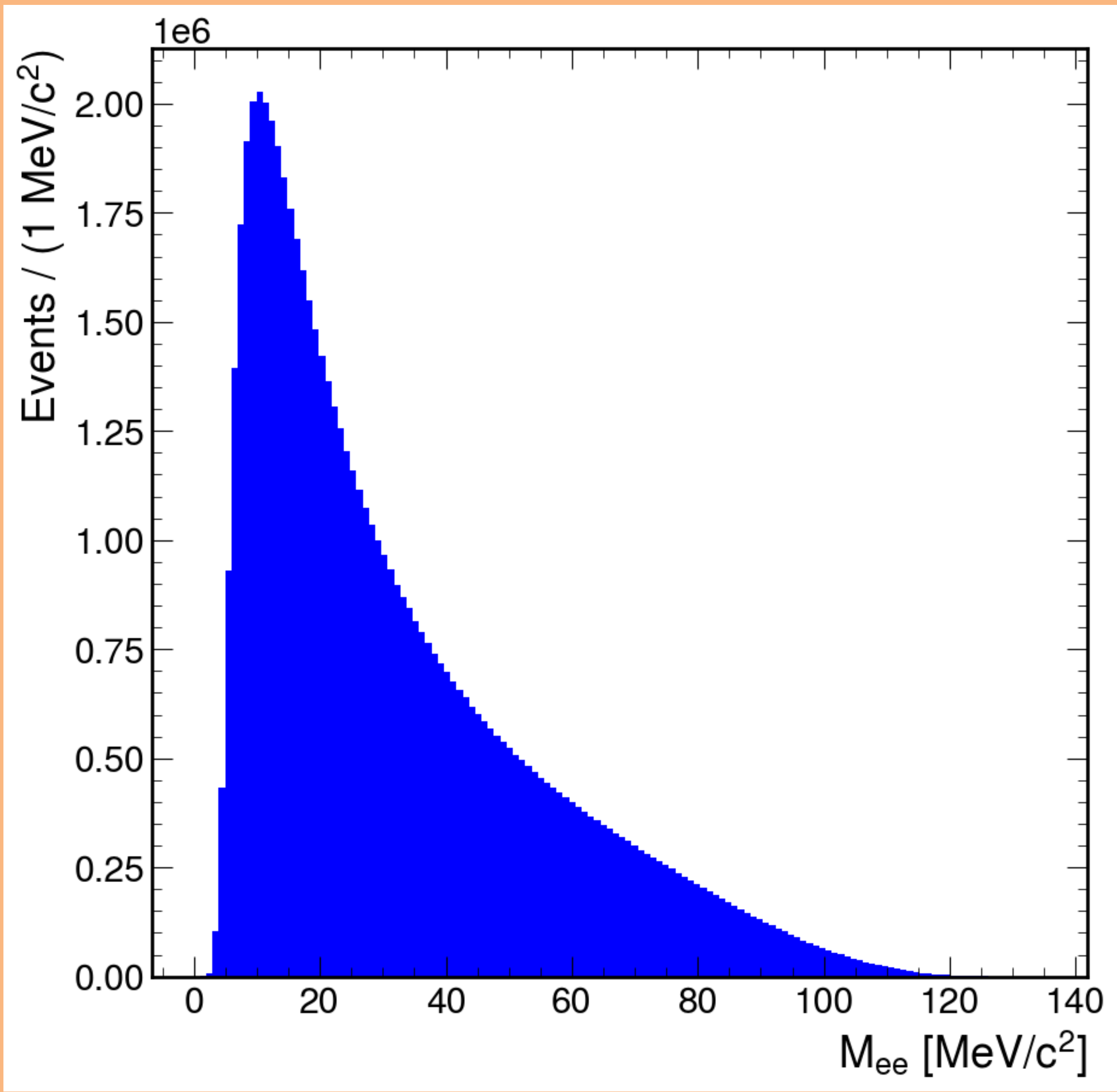

Figure 19: Reconstructed invariant mass of electron-positron pair for events passing the selection from the 2022 and 2023 NA62 dataset.

The produce an estimation of the background the distribution from the data itself was used. To estimate the background the distribution was fitted using polynomials using the side bands of the data and the process is described below.

A method similar to the previous section using the reconstructed mass resolution, defined with equation 0.12 was used to convert bins of M_{ee} to DP candidate windows. A new DP mass range was defined as $8 \leq m_{A'} \leq 121.6 MeV/c^2$. The mass ranges were extended compared to the previous MC approach due to the large statistical afforded by only using the data sample rather than the MC. Each DP mass hypothesis was defined by steps of $1\sigma_m(m_{A'})$. A 4th order polynomial was fitted to the data distribution using a least χ^2 method. The fitting region for each mass hypothesis was defined as $\pm 20\sigma_m(m_{A'})$ from the mass hypothesis and the distribution was re-binned for each with a bin width of $1/10\sigma_m(m_{A'})$. However data events in the mass bins above $125 MeV/c^2$ were removed due to the small counts causing instability with the fitting. Counts in the mass bins $\pm 1.5\sigma_m(m_{A'})$ of the mass hypothesis were excluded from the fit to avoid the possibility of sensitivity reduction due to the presence of a DP signal, which would result in a biased background estimate. The number of expected events N_{epx} was evaluated as the integral of the fitted functions across the $\pm 1.5\sigma_m(m_{A'})$ mass window divided the resultant bin width. The error on the number of expected events δN_{exp} was a combination of statistical and systematic. The statistical δN_{exp}^{stat} was computed from the propagation of the statistical errors of the polynomial parameters, provided in the form of a covariance matrix and was calculated using an error propagation method available in the ROOT package [**?**] (TF1::IntegralError()). The systematic error δN_{exp}^{sys} was estimated by fitting each distribution defined above with alternative polynomials, in this case 3rd and 5th order. The resultant integrals were evaluated and compared to the integral calculated using the main polynomial.

Example fits for background estimation and the systematic error estimation the first DP mass hypothesis is shown in Figure 20. The mass range of the plots is the entire $\pm 20\sigma_m(m_{A'})$ fitting range and shown the removed region $\pm 1.5\sigma_m(m_{A'})$ of the mass hypothesis described above.

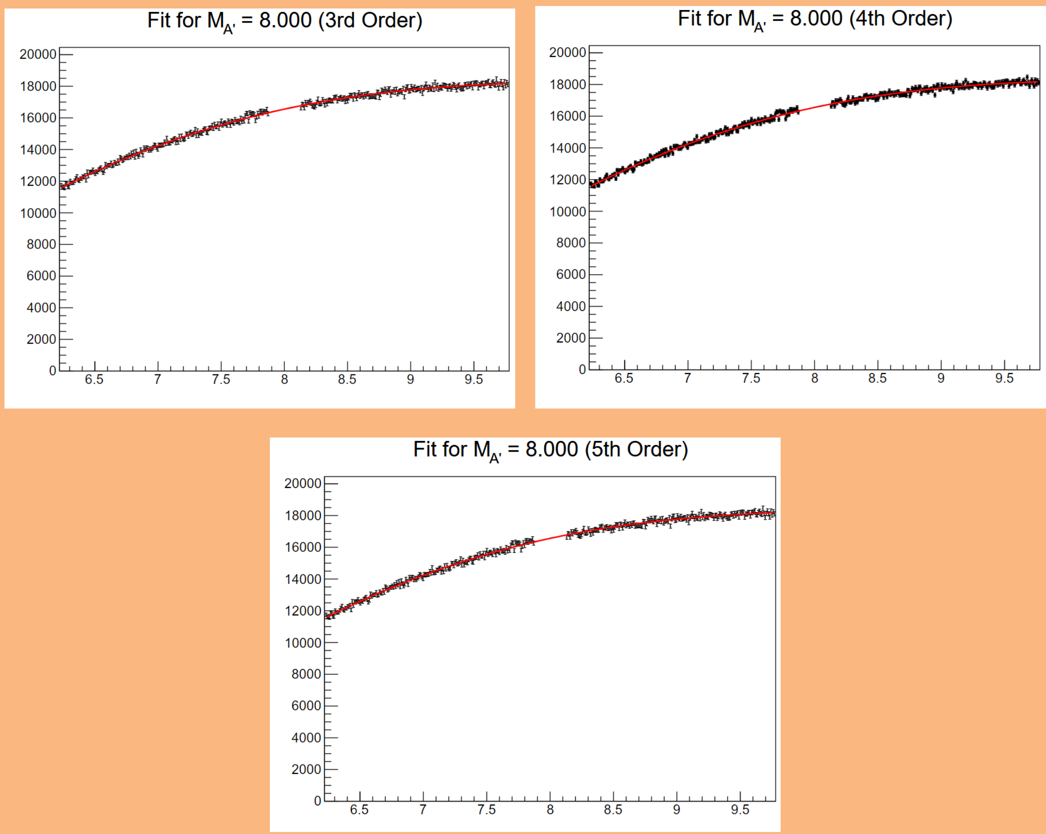

Figure 20: Fit of 4th order polynomial on the data distribution at the first DP mass hypothesis for the background estimation, along with fits for 3rd and 5th order polynomials used for systematic error estimation.

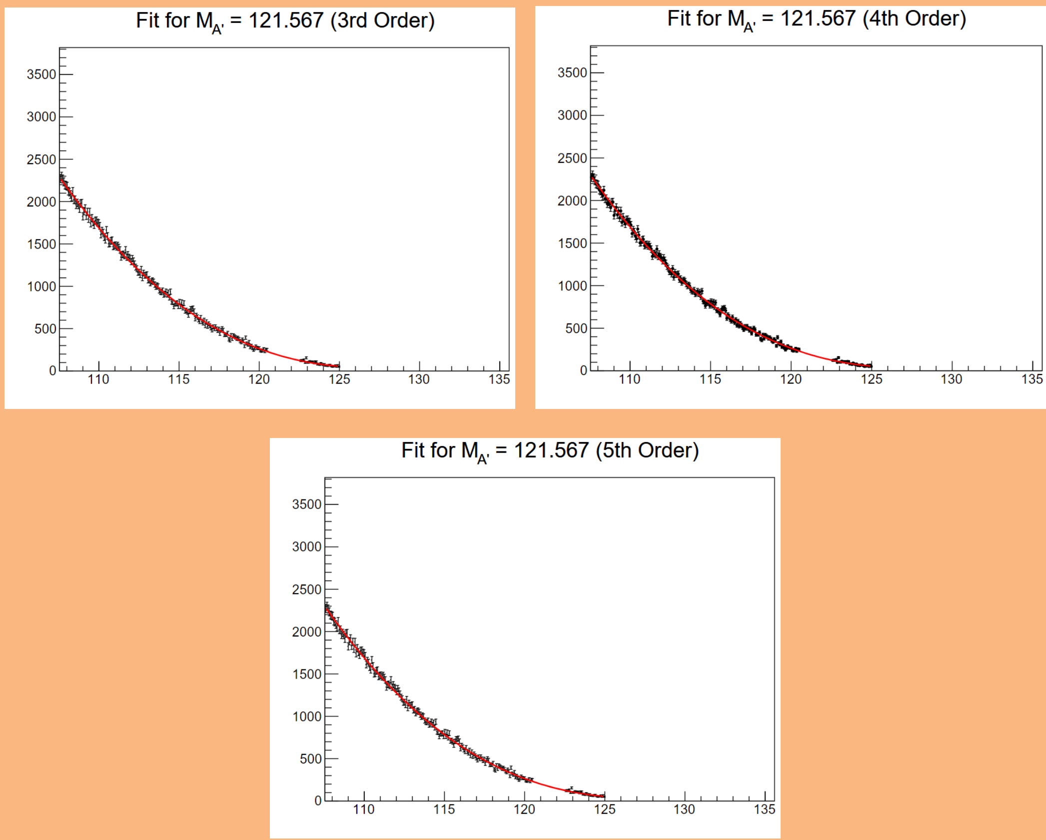

Figure 21: Fit of 4th order polynomial on the data distribution at the final DP mass hypothesis for the background estimation, along with fits for 3rd and 5th order polynomials used for systematic error estimation.

The fit for the final DP mass hypothesis is shown in Figure **??**, showing how all events after $125MeV/c^2$ have been removed from the fitting region. The figure shows that the upper limit was reaching a point where there was not many data points on the right side of the mass window, which will of inevitably increased the error in this region.

The resultant N_{obs}, N_{exp}, δN_{obs} and δN_{exp} are shown in Figure 22. As expected the combined expected error δN_{obs}, has dramatically reduced compared to the MC background presented in Figure 15 and out preforming the statistical error for the observed in each DP mass window. A described above the error δN_{obs} increases past $100MeV/c^2$ caused by the reduced fitting window due to the remove data events past

$125 MeV/c^2$, resulting in both the statistical and the systematic to increase. However this will not cause any significant impact to the analysis due to the competitive acceptance at this high mass range. Using the same method used in the MC study, N_{obs}, N_{exp} and δN_{exp} were used as inputs to the CLs method to calculate the upper limit on the number of DP candidates in each mass window and is shown in Figure 22. The resultant upper limit on the number of DP candidates was more than an order of magnitude better than the one produced by the Monte Carlo study and shown in Figure 15, however its not possible for a direct comparison due to the change in method and the increase in statistics from the added 2023 sample.

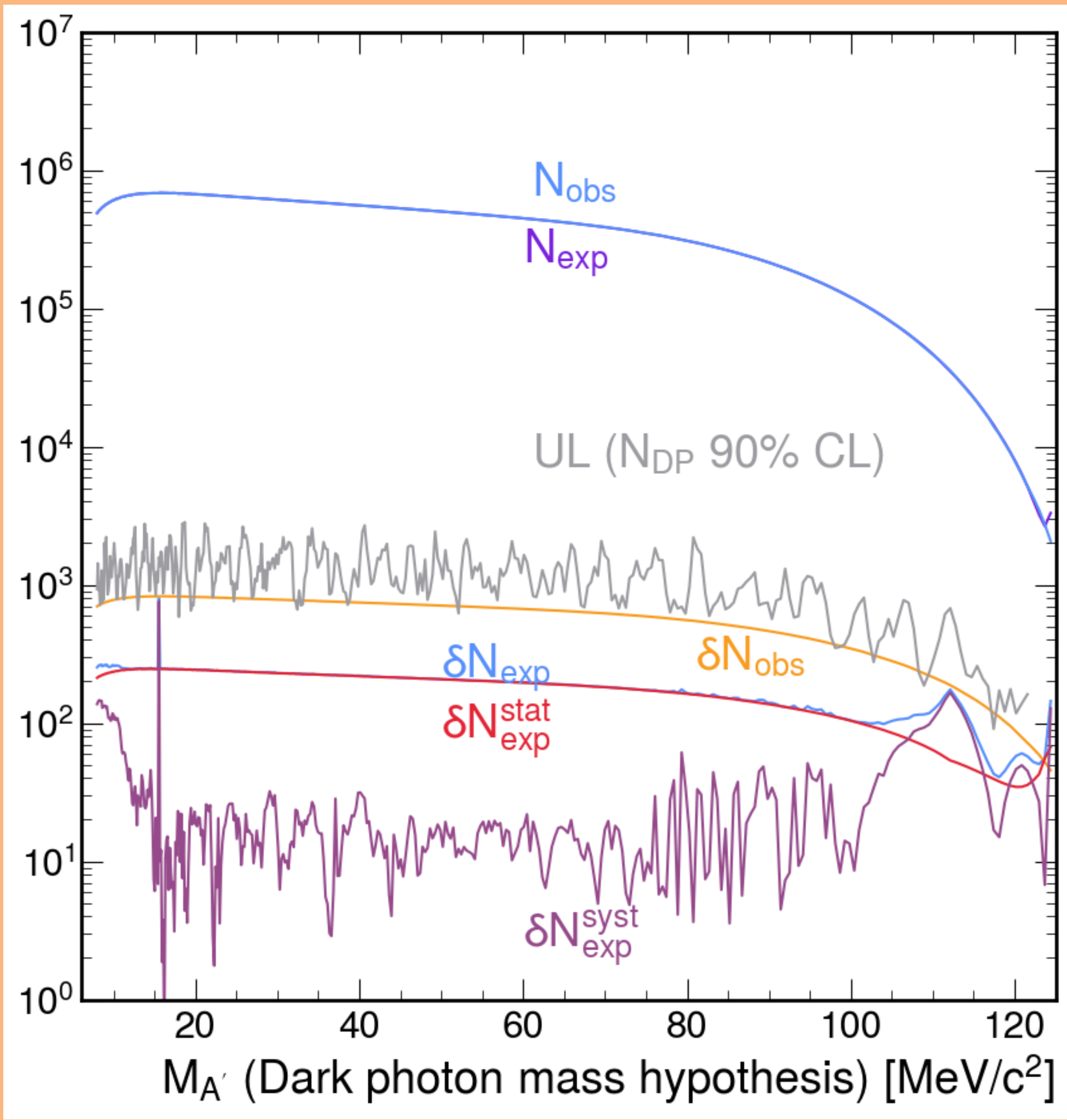

Figure 22: Number of observed data events (N_{obs}), expected number of events from the data driven method (N_{exp}), the respective errors (δN_{obs} and δN_{exp}) and the resultant upper limit on the number of Dark Photon candidates to a 90% confidence level.

With N_{obs}, N_{exp}, δN_{obs} and δN_{exp} shown in Figure 22, the local significance of the DP signal was calculated with Equation 0.13 and is shown in Figure 23. Again, the significance did not reach the threshold of three sigma so no DP was observed.

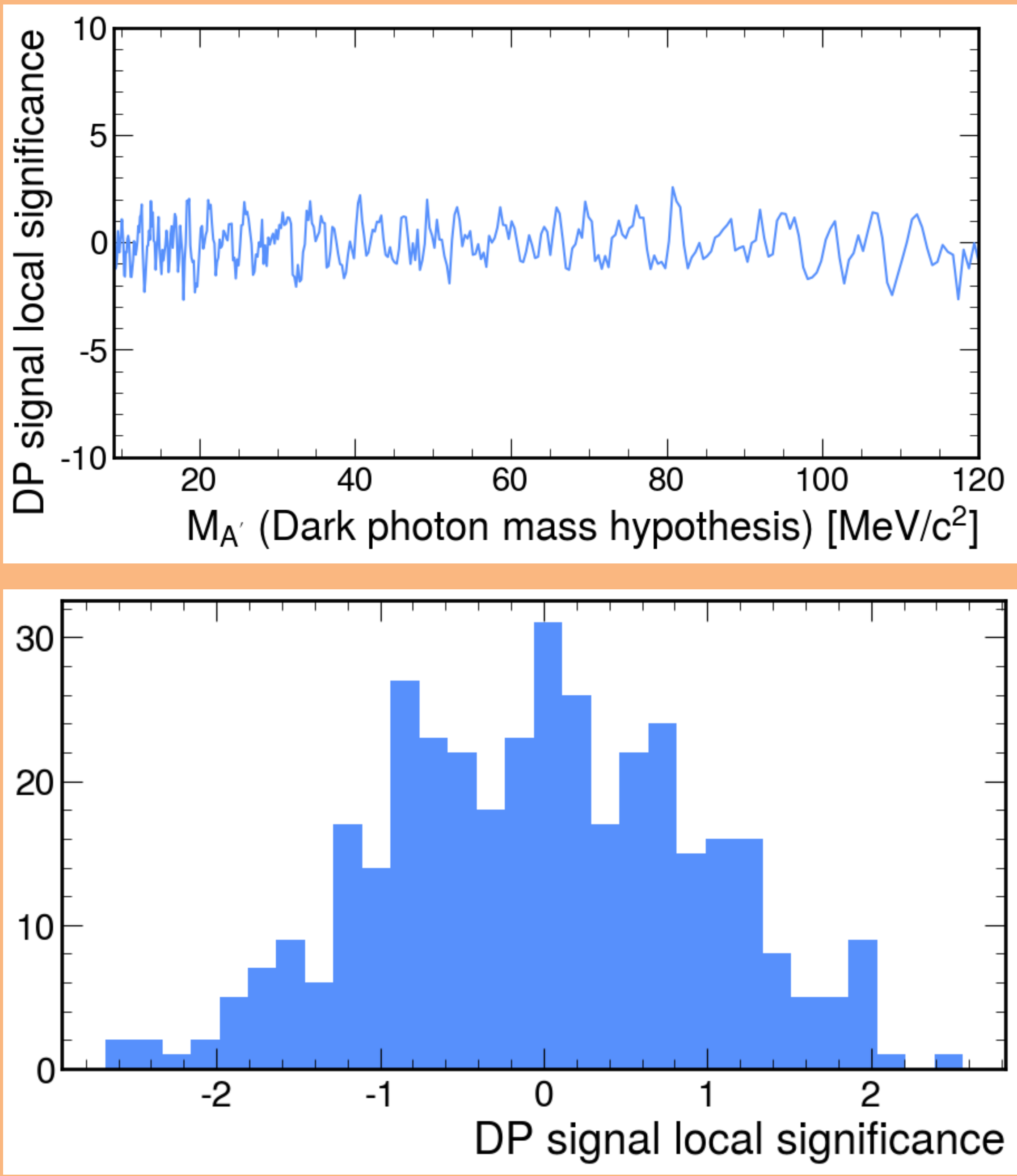

Figure 23: Local significance of DP signal across the search window $8 \leq m_{A'} \leq 121.6 MeV/c^2$ for the data driven study. Top, in terms of DP mass and bottom, the y projection of the DP mass plot.

The upper limit on the number of DP shown in Figure 22 was used to calculate the upper limit of the branching ratio to a 90% CL, using Equation 0.14 and is shown in Figure 24. The resultant upper limit for the $\mathcal{B}\pi \to \gamma A$ suggested a significant improvement compared to the result produced in the MC study shown in Figure 17.

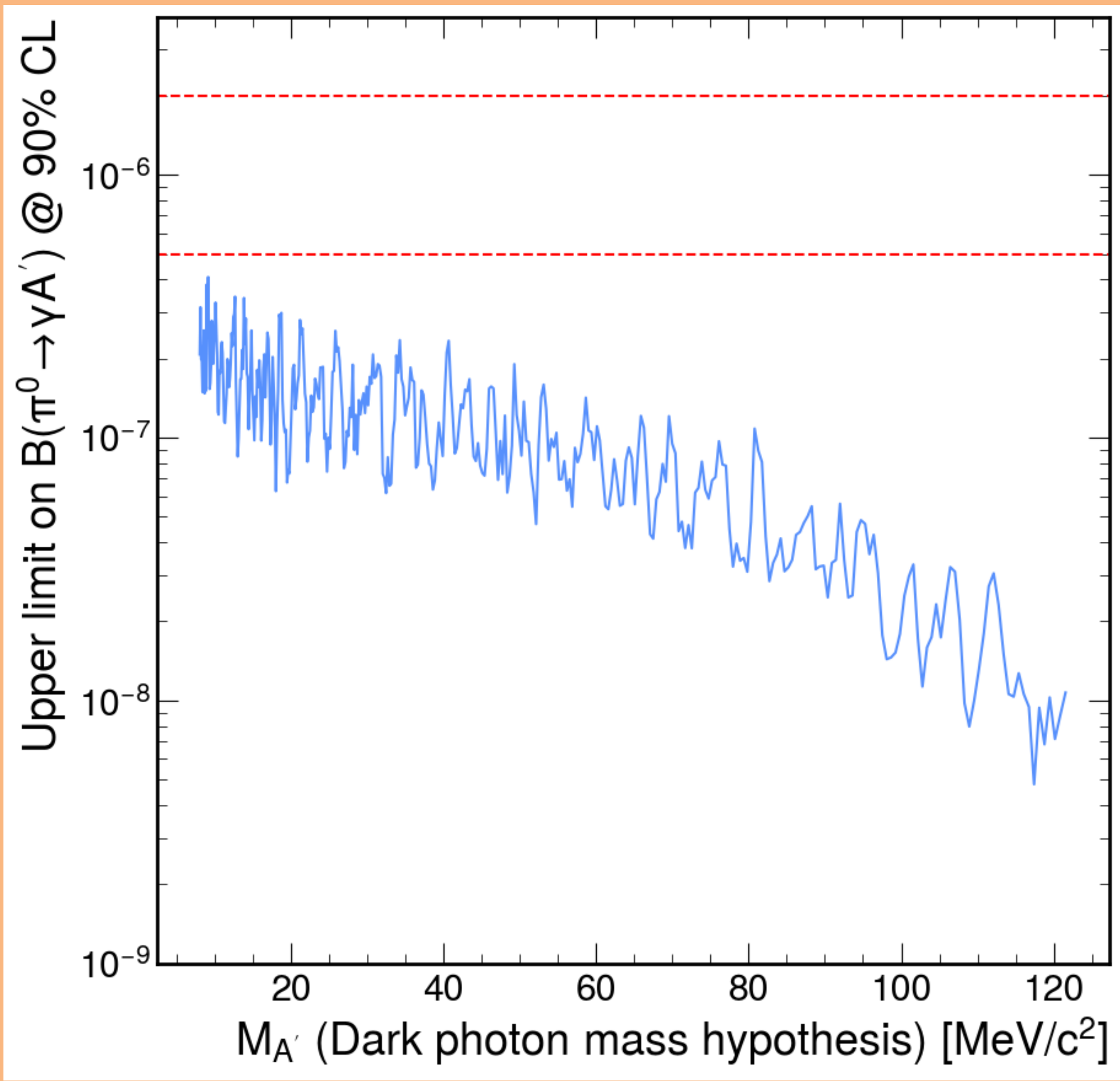

Figure 24: Upper limit on the branching fraction of $\pi^0 \to \gamma A'$ at a 90% CL for each Dark Photon mass value, for the data driven background method for estimating the number of expected events.

The resultant upper limit on the mixing parameter was calculated with Equation 0.15 was compared to the result from NA48/2 [**?**] in Figure 25.

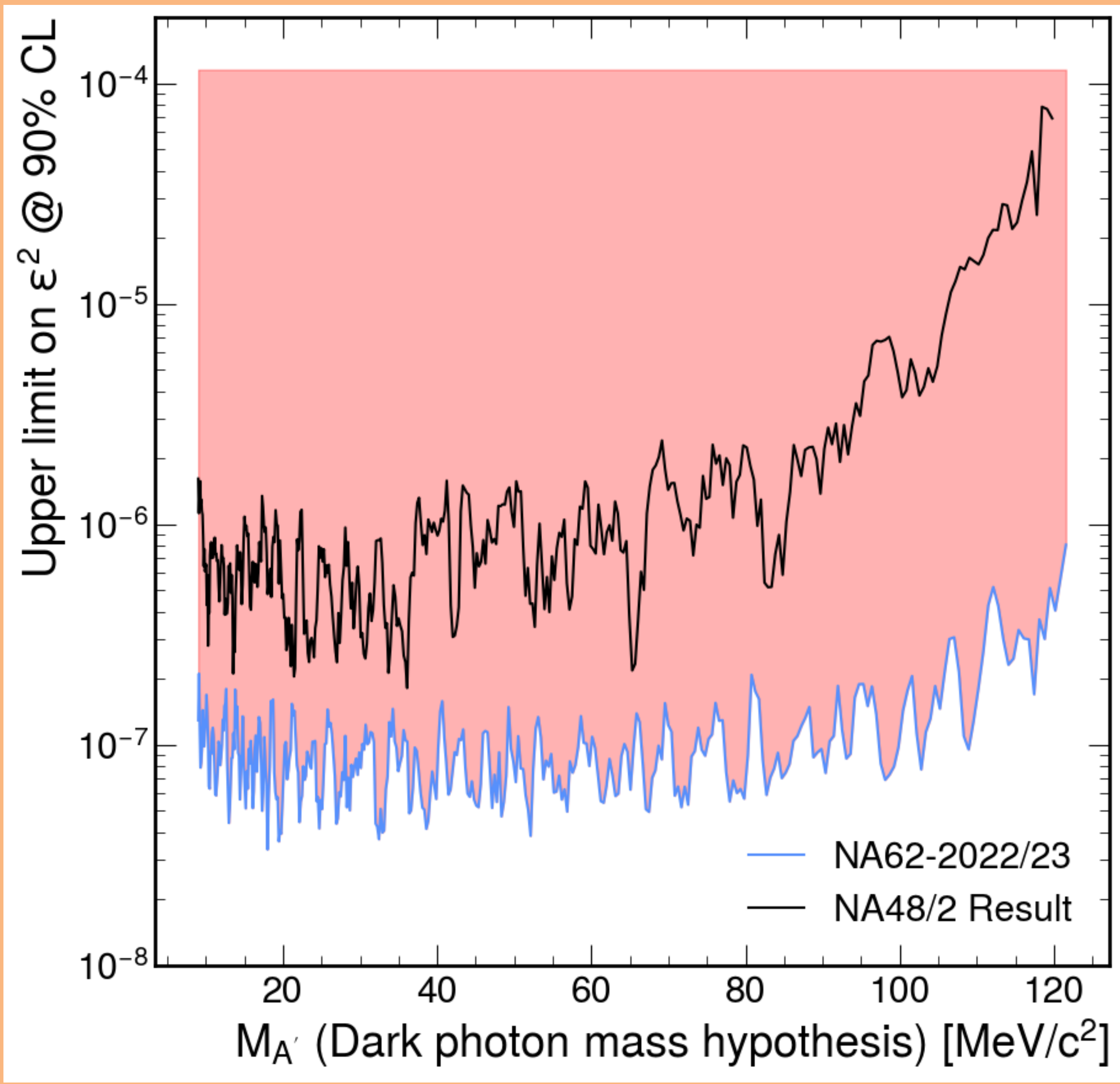

Figure 25: Upper limit mixing parameter ϵ^2 at a 90% CL for each Dark Photon mass value from the data driven background estimation, with the result from NA48/2 for comparison [**?**].

The study resulted in a significant improvement of the limit square mixing parameter compared to the NA48/2 result, by an order of magnitude. This was due in part to the much higher statistics provided by NA62 and the improved method for using the data for the estimation of the background, reducing the errors which limited the study previously. To see how this result compares to global results for the mixing parameter, the result from this analysis was presented in Figure 26, produced using the software package DarkCast, a companion software package to well known papers [? ?]. Figure 26 shows the result from this study and a number of other collaborations: NA48/2 [?], BaBar [?], NA64 [?], KLOE [?], KLOE-2 [?], FASER [?] and others [? ?]

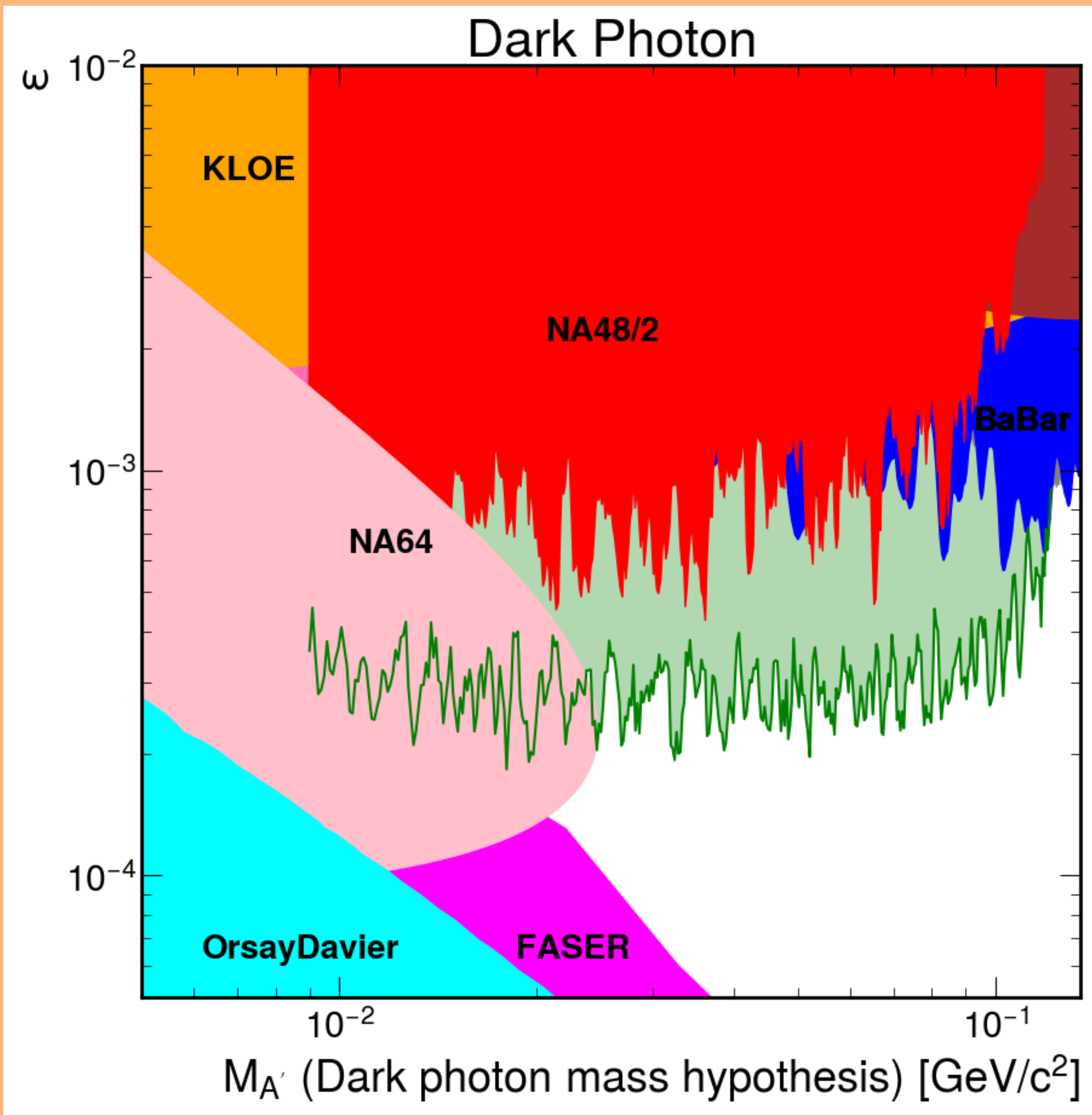

Figure 26: Upper limit at a 90% CL on the mixing parameter for the data driven background estimate (Green) compared to published exclusions limits [? ? ? ? ? ? ? ?]